CSE 276: ARTIFICIAL INTELLIGENCE FOUNDATIONS

Material Code Harmonisation Across Public Enterprises

AI-Driven Precision Entity Resolution & Duplicate Detection for Public Procurement

Section: K3P25LH | Group: G6 | Course: CSE 276 (AI Foundations)

Team Members (Draft / Allocation Record):

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Problem Source: Smart India Hackathon (SIH 2026) | Problem ID: SIH26099 | Ministry of Petroleum & Natural Gas

GitHub Repository: https://github.com/SHIVA07605/CSE-276-PROJECT.git (branch: ca1-trained-model)

Slide 2: Problem Context

The Industrial Problem: CPSEs (IOCL, OIL, NTPC) operate isolated ERP databases with bespoke codification.

Duplicate Spare Purchasing: Identical industrial equipment procured independently.

Beneficiaries: CPSE procurement and inventory managers, plant maintenance engineers, national exchequer.

Slide 3: Project Origin

SIH 2026 Problem Statement SIH26099: AI-Driven Standardization of Material Codes Across CPSEs.

Syllabus Units: Unit V NLP (normalization, embeddings); Unit VI Clustering (conflict-aware graph clustering); Unit V Evaluation (precision, recall, F1, purity).

Extension: Active learning uncertainty sampling.

Slide 4: Project Objectives

Objective 1: Build matching system proposing harmonised codes with evidence (374 multi-record clusters delivered).

Objective 2: Produce entity resolution model reporting pairwise precision/recall over labelled sample.

Objective 3: Report metrics vs exact match baseline, cluster purity, and Reviewer UI.

Slide 5: Dataset & Preprocessing

21,513 raw records -> 6,634 candidate pool across IOCL (41.4%), NTPC (32.0%), OIL (26.6%).

TF-IDF candidate blocking prunes 22.0 million pairs down to 22,464 candidate pairs (99.9% reduction).

Canonical spec parsing standardizes units and extracts technical attributes.

Slide 6: Methodology

Self-Supervised Learning by Construction (Ditto/Sudowoodo paradigm):

- 6,599 Synthetic Positives (format-only rewrites)

- 1,812 Synthetic Hard Negatives (1 spec mutated)

- 807 Real Spec Conflicts & 1,344 Cross-Commodity Negatives

Zero-leakage near-duplicate disjoint splits; 173 review records strictly excluded.

Slide 7: Implementation

34 hybrid features into HistGradientBoosting classifier.

Precision-first policy: Auto-merge (p >= 0.85 + shared specs), Auto-reject (p <= 0.25 or conflict), Needs Review (ambiguous).

Conflict-aware clustering blocking transitive conflicts; Streamlit Reviewer UI with explicit Save & Next.

Slide 8: Experimental Results

Held-Out Test (n=1,508): F1 0.994 vs Exact Match 0.165 (+0.829), Recall 0.989 vs 0.090, PR-AUC 0.9998.

Cluster Purity / Inverse Purity: 0.999 / 0.995 vs 1.000 / 0.545.

Review-Set Binary Labels (n=19): Acc 100%, 95% CI [0.83, 1.00].

Batch 2 Tier (n=25): 8 auto-merged with 100% precision, 13 auto-rejected with 100% NPV.

Slide 9: Demo & Verification

Live Reviewer UI with source provenance, conflict banners, and explicit save.

Live match_descriptions API with instant specification conflict detection.

197 automated test cases passing with zero regressions across 11 test modules.

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Slide 10: Limitations & Next Steps

Transparent Limitations: Held-out scores are an upper bound; review labels remain small in volume.

Active Learning Roadmap: batch2_review_priority.csv ranks pending pairs by uncertainty (|p - 0.5|).

Proposed Master Catalog: 374 clusters covering 913 records ready for ERP integration.